

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 5

A formula is given below for the n^{th} term a_n of a sequence $\{a_n\}$. Find the values of a_1 , a_2 , a_3 , and a_4 .

$$a_n = \frac{(-1)^{n+1}}{10n-5}$$

To find a_1 , substitute 1 for n into the given formula and evaluate.

$$\begin{aligned} a_1 &= \frac{(-1)^{(1)+1}}{10(1)-5} \\ &= \frac{1}{5} \end{aligned}$$

To find a_2 , substitute 2 for n into the given formula and evaluate.

$$\begin{aligned} a_2 &= \frac{(-1)^{(2)+1}}{10(2)-5} \\ &= -\frac{1}{15} \end{aligned}$$

To find a_3 , substitute 3 for n into the given formula and evaluate.

$$\begin{aligned} a_3 &= \frac{(-1)^{(3)+1}}{10(3)-5} \\ &= \frac{1}{25} \end{aligned}$$

To find a_4 , substitute 4 for n into the given formula and evaluate.

$$\begin{aligned} a_4 &= \frac{(-1)^{(4)+1}}{10(4)-5} \\ &= -\frac{1}{35} \end{aligned}$$